EXACT GRAM INVERSION FOR FRACTIONAL-PART APPROXIMATION

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Abstract. We solve the finite least-squares problem for ordinary sawtooth dilates by explicit Gram inversion and obtain its exact error and asymptotic constant. For the distinct reciprocal-dilate problem, we derive rational Gram certificates, rigorous residual intervals and Hardy-space zero bounds. The Riemann hypothesis remains open.

1. Three dictionaries and three different problems

Fractional-part approximation is sensitive to the placement of its variable.[1] On $(0,1)$, the functions $\{x/n\}$, $\{nx\}$ and $\{1/(nx)\}$ generate different spaces. Confusing them changes the optimization problem and can invalidate every later Mellin transform. This revision separates those spaces before doing any calculation.

Let $\{y\} = y - \lfloor y \rfloor$, and use the real Hilbert space $L^2(0,1)$ unless a complex transform is stated. For $M \geq 1$ define

$$r_n(x) = \{x/n\}, \qquad u_n(x) = \{nx\}, \qquad 1 \leq n \leq M.$$

For the reciprocal problem use the centered functions

$$g_n(x) = \{1/(nx)\} - \frac{1}{n}\{1/x\}, \qquad 2 \leq n \leq M. \tag{1}$$

Write E_M for the distance of 1 from $\operatorname{span}(u_1, \ldots, u_M)$, and D_M for the distance of 1 from $\operatorname{span}(g_2, \ldots, g_M)$. The notation keeps their limits distinct.

Theorem 1.1. *Let $J_2(n) = n^2 \prod_{p \mid n}(1 - p^{-2})$, with $J_2(1) = 1$, and put*

$$S_M = \sum_{n \leq M} \frac{\varphi(n)^2}{J_2(n)}.$$

The ordinary-dilate problem has the unique solution

$$E_M^2 = \frac{1}{1 + 3S_M}, \tag{2}$$

$$c_j^{(M)} = \frac{6j}{1 + 3S_M} \sum_{\substack{k \leq M \\ j \mid k}} \mu(k/j) \frac{\varphi(k)}{J_2(k)}, \quad 1 \leq j \leq M. \tag{3}$$

Moreover

$$E_M^2 \sim \frac{1}{3C_0 M}, \qquad C_0 = \prod_p \left(1 - \frac{2}{p(p+1)}\right) > 0. \tag{4}$$

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[1]Changing a function at its countably many jump points does not change its L^2 class, but exchanging nx and $1/(nx)$ changes the class on a set of positive measure.

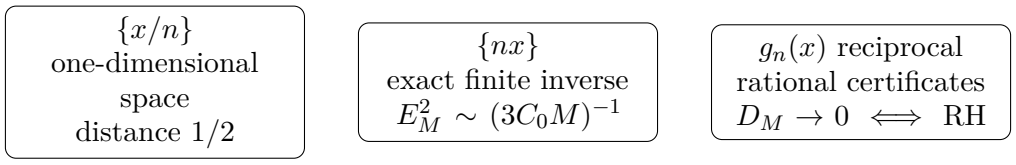


FIGURE 1. The three dictionaries. The ordinary-dilate asymptotic is a proved approximation theorem. The reciprocal limit is the RH criterion.

All these statements are unconditional. They concern the functions $\{nx\}$ and do not prove a limit statement for D_M.

The exact Gram matrix behind the theorem is a classical GCD matrix plus a rank-one term. Smith's determinant factorization [1] and the broader theory of GCD sums [2] provide its historical setting. We give the factorization, inverse and optimization proofs explicitly rather than treating a floating-point inverse as a proof.

For the reciprocal functions, the strong Nyman–Beurling criterion of Báez-Duarte [4], building on Beurling [3], gives

$$\text{RH} \quad \Longleftrightarrow \quad \lim_{M\to\infty} D_M = 0. \tag{5}$$

That equivalence is cited, not reproved. We derive finite certificates for this problem and explain exactly why they do not establish its unknown limit. The paper also corrects the Mellin, reproducing-kernel and filtering assertions in the previous version [6].

2. Rank-one collapse of the linear dictionary

For $0 < x < 1$ and $n \geq 1$, one has $0 < x/n < 1$ and therefore $r_n(x) = x/n$. Every finite span is the line generated by x. Projection of 1 onto that line gives

$$\frac{\langle 1, x\rangle}{\langle x, x\rangle} x = \frac{3}{2}x, \qquad \left\| 1 - \frac{3}{2}x \right\|_2^2 = 1 - \frac{(1/2)^2}{1/3} = \frac{1}{4}.$$

Thus its distance is $1/2$, independently of M.

Writing $d = (1, 1/2, \ldots, 1/M)^T$, its Gram matrix is $G^{\text{lin}} = dd^T/3$. It has one nonzero eigenvalue $\|d\|^2/3$, and its Moore–Penrose inverse is

$$(G^{\text{lin}})^+ = \frac{3dd^T}{\|d\|^4}.$$

Indeed both matrices act by the reciprocal eigenvalues on $\operatorname{span}(d)$ and vanish on its orthogonal complement. This proves the four Moore–Penrose identities directly.

Remark 2.1. Increasing the number of coefficients in this dictionary cannot add directions to the approximation space. A numerical routine returning different fitted functions at different M is then measuring a conditioning or implementation effect, not a change in the exact span.

3. The ordinary sawtooth Gram matrix

3.1. **An exact inner product.** Set $v_n = u_n - 1/2$. Its mean is zero. The L^2 Fourier expansion of the centered sawtooth is

$$\{x\} - \frac{1}{2} = -\sum_{k\geq 1} \frac{\sin(2\pi kx)}{\pi k}.$$

This follows by integrating against each sine and cosine. Completeness of the trigonometric system gives the L^2 identity; pointwise values at jumps do not affect an inner product.

Lemma 3.1. *For positive integers j, k,*

$$\langle v_j, v_k\rangle = \frac{\gcd(j,k)^2}{12jk}, \qquad \langle u_j, u_k\rangle = \frac{1}{4} + \frac{\gcd(j,k)^2}{12jk}.$$

Proof. Write $j = dj'$, $k = dk'$ with $(j', k') = 1$. In the two Fourier series only matching frequencies contribute. The equation $\ell j = mk$ requires $\ell = k't$, $m = j't$, $t \geq 1$. Orthogonality therefore gives

$$\langle v_j, v_k\rangle = \frac{1}{2\pi^2 j'k'} \sum_{t\geq 1} \frac{1}{t^2} = \frac{1}{12j'k'}.$$

The matching-frequency series is absolutely convergent, justifying the pairing of the L^2 expansions. Adding the product of the means gives the second formula. □

This formula has no unresolved correction term. For example $G_{11} = 1/3$, $G_{12} = 7/24$ and $G_{22} = 1/3$. An independent check in the accompanying program integrates each piecewise quadratic product over the union of its rational breakpoints.

3.2. **Divisibility factorization.** Let Z be the lower triangular matrix $Z_{jd} = \mathbf{1}_{d|j}$, indexed by $1 \leq j, d \leq M$, and let $D = \operatorname{diag}(1, 2, \ldots, M)$. The identity $\sum_{d|n} J_2(d) = n^2$ follows first for prime powers by telescoping and then for general n by multiplicativity. Hence

$$H_{jk} := \gcd(j,k)^2, qquadH = Z \operatorname{diag}(J_2(1), \ldots, J_2(M)) Z^T. \tag{6}$$

The inverse of Z has entries

$$(Z^{-1})_{jd} = \begin{cases} \mu(j/d), & d \mid j, \\ 0, & d \nmid j. \end{cases}$$

Multiplication proves this formula because $\sum_{d|n} \mu(d)$ is one for $n = 1$ and zero otherwise.

If C is the centered Gram matrix, Lemma 3.1 gives

$$C = \frac{1}{12} D^{-1} Z \operatorname{diag}(J_2) Z^T D^{-1}, \tag{7}$$

$$C^{-1} = 12 D Z^{-T} \operatorname{diag}(J_2^{-1}) Z^{-1} D. \tag{8}$$

All $J_2(n)$ are positive, so C is positive definite. Moreover

$$\det C = \frac{\prod_{n\leq M} J_2(n)}{12^M (M!)^2}. \tag{9}$$

The uncentered matrix is $G = C + \mathbf{1}\mathbf{1}^T/4$, and is also positive definite. In particular the finite minimizer is unique.

4. Exact inverse and least-squares solution

Put $w = C^{-1}\mathbf{1}$ and $A_M = \mathbf{1}^T w$. Direct multiplication yields

$$G^{-1} = C^{-1} - \frac{w w^T}{4 + A_M}. \tag{10}$$

For completeness, multiplying the right side by $C + \mathbf{1}\mathbf{1}^T/4$ leaves I plus the coefficient $1/4 - 1/(4 + A_M) - A_M/[4(4 + A_M)] = 0$ times $\mathbf{1}w^T$. Thus no matrix-inversion theorem beyond multiplication is required.

Let $q = (1, 2, \ldots, M)^T = D\mathbf{1}$. The divisor identity $\sum_{d|n} \mu(n/d)d = \varphi(n)$ gives $Z^{-1}q = (\varphi(1), \ldots, \varphi(M))^T$. Therefore

$$A_M = 12 \sum_{n\leq M} \frac{\varphi(n)^2}{J_2(n)} = 12 S_M. \tag{11}$$

The jth entry of w is

$$w_j = 12j \sum_{\substack{k \le M \\ j \mid k}} \mu(k/j) \frac{\varphi(k)}{J_2(k)}. \tag{12}$$

Proof of the exact formulas in Theorem 1.1. The target vector is $b_j = \langle 1, u_j \rangle = 1/2$. The normal equations are $Gc = b = \mathbf{1}/2$. Equations (10) and (11) give

$$c = \frac{w}{2(1 + 3S_M)},$$

which, by (12), is (3). Completing the quadratic square in $1 - 2b^T c + c^T G c$ gives

$$E_M^2 = 1 - b^T G^{-1} b = 1 - \frac{3S_M}{1 + 3S_M} = \frac{1}{1 + 3S_M}.$$

This also proves optimality directly. The determinant identity

$$\det G = \det C \, (1 + 3S_M)$$

follows by factoring out C and observing that a rank-one matrix has only one possibly nonzero eigenvalue. □

For $M = 1$, $S_1 = 1$, $c_1 = 3/2$ and $E_1^2 = 1/4$. For $M = 2$, $S_2 = 4/3$, the coefficients are $(4/5, 4/5)$ and $E_2^2 = 1/5$. These checks also distinguish this model from the rank-one family of the preceding section.

4.1. **An arbitrary target.** For real $f \in L^2(0,1)$ put $b_j = \langle f, u_j \rangle$. The unique coefficient vector is

$$c = C^{-1} b - \frac{w(w^T b)}{4 + 12S_M},$$

and the squared error is

$$\|f\|_2^2 - b^T C^{-1} b + \frac{(w^T b)^2}{4 + 12S_M}.$$

The complex case uses conjugate transposes. Formula (8) expresses every matrix entry by finite divisor sums. Thus once the target inner products are known, the optimization requires no uncontrolled numerical inversion.

4.2. **Arbitrary finite divisor-closed sets.** An initial interval of integers is not essential to the exact inverse.

Theorem 4.1. *Let $\Lambda \subset \mathbb{N}$ be finite, nonempty and closed under taking positive divisors. Put $S_\Lambda = \sum_{n \in \Lambda} \varphi(n)^2 / J_2(n)$. Then*

$$\operatorname{dist}\big(1, \operatorname{span}\{u_n : n \in \Lambda\}\big)^2 = \frac{1}{1 + 3S_\Lambda}.$$

The unique minimizing coefficients are

$$c_j = \frac{6j}{1 + 3S_\Lambda} \sum_{\substack{k \in \Lambda \\ j \mid k}} \mu(k/j) \frac{\varphi(k)}{J_2(k)}, \qquad j \in \Lambda.$$

If $\Lambda \subset \Gamma$ are two such sets, the exact improvement in squared error is

$$\frac{3(S_\Gamma - S_\Lambda)}{(1 + 3S_\Lambda)(1 + 3S_\Gamma)}.$$

Proof. Order Λ increasingly and restrict D and Z to those indices. Every divisor of a common divisor of two indices remains in Λ, so the factorization (6) is unchanged. Every interval of the divisibility order between two members of Λ contains all its integer divisors. Its incidence inverse is therefore still $\mu(j/d)$ when $d \mid j$. Finally $Z^{-1}(j)_{j\in\Lambda} = (\varphi(j))_{j\in\Lambda}$ because all divisors of j are present. The proof of (2)–(3) now applies verbatim. Subtracting the two reciprocal error formulas proves the last identity. □

For example $\Lambda = \{1, 2, 3, 6\}$ has $S_\Lambda = 2$ and squared error $1/7$. Such a divisor-closed set need not be an initial interval. The theorem does not apply to an arbitrary subset without its divisors: the restricted incidence inverse would then need to be recalculated.

5. The asymptotic constant and coefficient decay

Define $h(n) = \varphi(n)^2/J_2(n)$. At prime powers,

$$h(p^k) = \frac{p-1}{p+1} \quad (k \geq 1).$$

Write $h = 1 * g$ in Dirichlet-convolution notation. Then g is multiplicative, vanishes on nonsquarefree integers and satisfies

$$g(p) = -\frac{2}{p+1}, \qquad |g(n)| \leq \frac{\tau(n)}{n}.$$

Since $\sum \tau(n)/n^2 < \infty$, the series $\sum g(n)/n$ converges absolutely and equals the Euler product C_0 of (4). Every factor is positive and the sum of their defects is finite, so $C_0 > 0$.

Proposition 5.1. *For every integer $M \geq 1$,*

$$|S_M - C_0 M| \leq (1 + \log M)^2 + 2\log M + 4. \tag{13}$$

Furthermore $C_0 \geq \zeta(2)^{-2} = 36/\pi^4$.

Proof. Divisor summation gives

$$S_M = \sum_{d\leq M} g(d)\lfloor M/d \rfloor.$$

Replacing the floor by M/d incurs an error at most

$$\sum_{d\leq M} |g(d)| \leq \sum_{d\leq M} \frac{\tau(d)}{d} \leq \left(\sum_{j\leq M} \frac{1}{j}\right)^2 \leq (1 + \log M)^2.$$

For the omitted tail use $T(t) = \sum_{n\leq t} \tau(n) \leq t(1 + \log t)$. Partial summation and deletion of its nonpositive endpoint term give

$$M \sum_{d>M} \frac{\tau(d)}{d^2} \leq 2M \int_M^\infty \frac{1 + \log t}{t^2}\, dt = 2\log M + 4.$$

This proves (13). Finally, for each prime p,

$$1 - \frac{2}{p(p+1)} \geq (1 - p^{-2})^2,$$

as is seen by subtraction. Multiplying finite products and taking limits proves the stated lower bound. □

The asymptotic in Theorem 1.1 follows immediately. It is a theorem about ordinary sawtooth approximation, not a conditional consequence of RH.

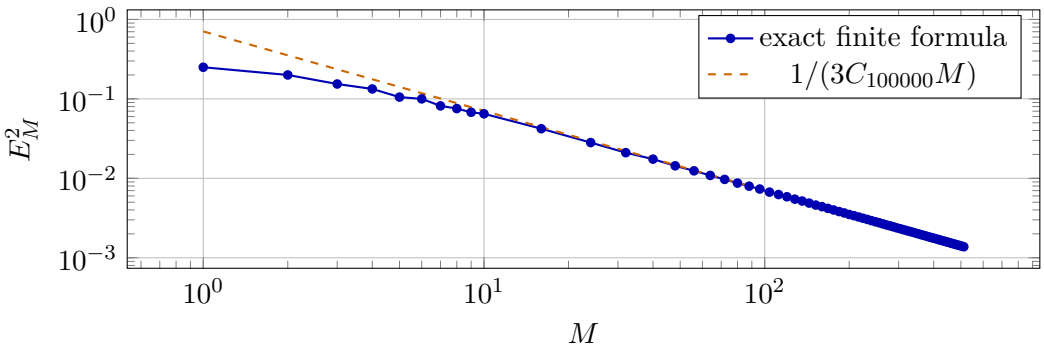


FIGURE 2. Finite errors from exact rational S_M. The comparison uses the explicitly truncated prime product C_{100000}, not an asserted decimal value for the infinite product. All plotted values are regenerated by the accompanying checker.

Proposition 5.2. *Let $H_k = \sum_{j\le k} 1/j$. Then*

$$|c_j^{(M)}| \le \frac{6H_{\lfloor M/j\rfloor}}{1+3S_M}, \qquad \sum_{j\le M} |c_j^{(M)}|^2 \le \frac{180M}{(1+3S_M)^2}.$$

In particular every fixed coefficient tends to zero and the squared coefficient norm is $O(M^{-1})$.

Proof. The prime-power formula gives $\varphi(k)/J_2(k) \le 1/k$. Taking absolute values in (3) yields the first bound. Since $H_{\lfloor M/j\rfloor} \le 1 + \log(M/j)$ and the latter squared is decreasing in j,

$$\sum_{j=1}^{M} (1+\log(M/j))^2 \le \int_0^M (1+\log(M/x))^2\,dx = 5M.$$

The second bound follows. Proposition 5.1 supplies the asymptotic conclusions. □

Thus the least-squares coefficients in this problem do not converge termwise to $\mu(j)$. The Möbius function enters the inverse of the divisibility matrix; its appearance there is not a termwise coefficient limit.

6. MELLIN TRANSFORMS WITH THE CORRECT DICTIONARY

For f on $(0,1)$ write $\mathcal{M}f(s) = \int_0^1 f(x)x^{s-1}\,dx$ wherever the integral exists. The elementary identity

$$\{nx\} = nx - \sum_{j=1}^{n-1} \mathbf{1}_{[j/n,1)}(x)$$

gives, for $\Re s > 0$,

$$\mathcal{M}u_n(s) = \frac{n}{s+1} - \frac{n-1}{s} + \frac{n^{-s}}{s}\sum_{j=1}^{n-1} j^s. \tag{14}$$

There is no zeta factor in this finite expression. Apparent singularities at zero are removable whenever the original integral continues there.

For the reciprocal functions let $0 < \theta \le 1$. When $\Re s > 1$, expand the floor into nonnegative indicator functions and integrate:

$$\int_0^1 \{\theta/x\} x^{s-1}\,dx = \theta \int_0^1 x^{s-2}\,dx - \sum_{k\ge 1} \int_0^{\theta/k} x^{s-1}\,dx$$
$$= \frac{\theta}{s-1} - \frac{\theta^s \zeta(s)}{s}.$$

For complex s absolute convergence justifies the sum; it follows first on the real axis and then analytically. The left side is analytic on $\Re s > 0$, since the fractional part is bounded. The right side has a removable singularity at $s = 1$ and agrees there by continuation. Consequently

$$\mathcal{M} g_n(s) = -\frac{\zeta(s)}{s}(n^{-s} - n^{-1}), \qquad \Re s > 0. \tag{15}$$

At $s = 1$, the zero of $n^{-s} - n^{-1}$ cancels the pole of ζ, and

$$b_n := \langle 1, g_n \rangle = \frac{\log n}{n}. \tag{16}$$

6.1. **Why the centered criterion is the right one.** The strong Báez-Duarte criterion is stated in $L^2(0,\infty)$ using $\rho_n(x) = \{1/(nx)\}$ and target $\mathbf{1}_{(0,1)}$ [4]. For $x > 1$, $\rho_n(x) = 1/(nx)$. If $f = \sum_{n\le M} a_n \rho_n$ and $A = \sum a_n/n$, then its squared tail norm on $(1,\infty)$ is A^2. Also $\rho_1 \in L^2(0,\infty)$. Thus any approximating sequence with error tending to zero has $A \to 0$, and replacing f by $f - A\rho_1$ changes its norm by at most $|A|\|\rho_1\|_2 \to 0$. The replacement equals $\sum_{n=2}^{M} a_n(\rho_n - n^{-1}\rho_1)$ and vanishes for $x > 1$. Conversely such centered approximants extend by zero and already lie in the original integer-dilate span. This proves equivalence between the cited criterion and (5).

7. Rational certificates for the reciprocal Gram matrix

7.1. **Step values and a positive series.** Let $q = \lfloor 1/x \rfloor$, so $1/x = q + t$ with $0 \le t < 1$. For any integer $n \ge 2$,

$$\lfloor (q+t)/n \rfloor = \lfloor q/n \rfloor.$$

Substitution in (1) gives the exact step-function identity

$$g_n(x) = \{q/n\} \quad \text{for } \frac{1}{q+1} < x \le \frac{1}{q}. \tag{17}$$

Boundary conventions affect a null set only.

Let $v_q = (\{q/2\}, \ldots, \{q/M\})^T$ and $w_q = 1/[q(q+1)]$. The reciprocal Gram matrix is

$$B = \sum_{q\ge 1} w_q v_q v_q^T, qquad B^{(Q)} = \sum_{q=1}^{Q} w_q v_q v_q^T. \tag{18}$$

Every truncated matrix is rational and positive semidefinite. The tail is positive semidefinite as well. Entrywise,

$$0 \le B_{nm} - B^{(Q)}_{nm} \le \sum_{q>Q} w_q = \frac{1}{Q+1}. \tag{19}$$

For a real vector c, its quadratic tail obeys

$$0 \le c^T (B - B^{(Q)}) c \le \frac{\|c\|_1^2}{Q+1}. \tag{20}$$

The lower bound uses positive semidefiniteness, not entrywise positivity with arbitrary signed coefficients. This distinction is important when using optimized vectors.

7.2. **A finite determinant certificate.**

Theorem 7.1. *For $M \geq 2$, let V have rows $q = 1, \dots, M-1$ and columns $n = 2, \dots, M$, with $V_{qn} = \{q/n\}$. Then*

$$|\det V| = \frac{1}{M}, \qquad \det B^{(M-1)} = \frac{1}{M(M!)^2}.$$

In particular $B^{(Q)}$ is positive definite for every $Q \geq M-1$, and so is B.

Proof. Retain the first row. For $q \geq 2$, replace row q by its difference with row $q-1$, performing these operations in descending order. The new entry is $1/n - \mathbf{1}_{n|q}$. Subtract the first row, whose entries are $1/n$, from each of these rows. The resulting lower rows have entries $-\mathbf{1}_{n|q}$. Their block with $2 \leq q, n \leq M-1$ is lower triangular with diagonal -1. The final column $n = M$ vanishes in every lower row and has first entry $1/M$. Expansion along this column proves the first identity.

Since $B^{(M-1)} = V^T \operatorname{diag}(w_1, \dots, w_{M-1}) V$,

$$\det B^{(M-1)} = \frac{1}{M^2} \prod_{q=1}^{M-1} \frac{1}{q(q+1)} = \frac{1}{M(M!)^2}.$$

All diagonal weights are positive and V is nonsingular. Adding further positive semidefinite terms preserves positive definiteness. □

A crude but completely explicit lower eigenvalue certificate follows. Put $N = M-1$. Since the diagonal entries of $B^{(M-1)}$ are at most one, its trace is at most N. All eigenvalues are positive, and the product of the $N-1$ largest is at most N^{N-1}. Hence

$$\lambda_{\min}(B) \geq \lambda_{\min}(B^{(M-1)}) \geq \frac{1}{M(M!)^2 N^{N-1}}. \tag{21}$$

For $N = 1$ the power N^0 is one. The bound is conservative, but requires neither floating-point diagonalization nor a conjecture.

7.3. **Certified residual intervals.** For any rational vector $c = (c_2, \dots, c_M)$ define

$$R(c) = \left\| 1 - \sum_{n=2}^{M} c_n g_n \right\|_2^2.$$

Equations (16) and (18) give

$$R(c) = 1 - 2 \sum_{n=2}^{M} \frac{c_n \log n}{n} + \sum_{q \geq 1} \frac{(c^T v_q)^2}{q(q+1)}. \tag{22}$$

Suppose rational intervals $L_n \leq \log n \leq U_n$ have been proved. Taking sign-aware interval bounds in the finite linear sum and using (20) then gives a rigorous interval for $R(c)$. The checker uses the positive series

$$\log x = 2 \sum_{k \geq 0} \frac{t^{2k+1}}{2k+1}, \qquad t = \frac{x-1}{x+1}, \quad x \geq 1. \tag{23}$$

After K terms the omitted tail is between zero and $2t^{2K+1}/[(2K+1)(1-t^2)]$. Thus all endpoints can be rational. Range reduction by powers of two keeps this certificate efficient.

These intervals certify individual finite residuals, and therefore upper bounds for D_M^2.[2] They do not certify that an unspecified sequence of residuals tends to zero. The determinant theorem proves finite independence, not the density assertion in (5).

[2]A finite upper bound is a proved inequality for chosen coefficients. Convergence to zero requires a uniform analytic estimate for a sequence of residuals.

8. The Mellin Hardy space and zero obstructions

8.1. **The correct reproducing kernel.** For $f \in L^2(0,1)$ set $F(s) = \mathcal{M}f(s)$ on $\Re s > 1/2$. Cauchy–Schwarz gives

$$|F(s)| \le \frac{\|f\|_2}{\sqrt{2\Re s - 1}}. \tag{24}$$

The kernel is

$$K(s,w) = \int_0^1 x^{s+\overline{w}-2}\,dx = \frac{1}{s+\overline{w}-1}. \tag{25}$$

Under $x = e^{-t}$ and $h(t) = e^{-t/2}f(e^{-t})$,

$$F(1/2+z) = \int_0^\infty h(t)e^{-zt}\,dt, \qquad \|h\|_{L^2(0,\infty)} = \|f\|_2.$$

Plancherel on vertical lines therefore gives the Hardy norm identity

$$\sup_{\sigma>1/2}\frac{1}{2\pi}\int_{\mathbb{R}}|F(\sigma+it)|^2dt = \|f\|_2^2.$$

The supremum is the limit as $\sigma \downarrow 1/2$, by monotone convergence applied to $|h(t)|^2e^{-2(\sigma-1/2)t}$. These formulas describe the Mellin image as a reproducing-kernel Hilbert space; the usual Laplace-transform Hardy-space theorem identifies it with $H^2(\Re s > 1/2)$. The norm and evaluations used here already follow from the displayed calculation.

Boundary point evaluation is not bounded. For fixed $t_0 \in \mathbb{R}$, let

$$f_N(x) = \frac{x^{-1/2-it_0}}{\sqrt{N}}\mathbf{1}_{[e^{-N},1]}(x).$$

Then $\|f_N\|_2 = 1$ but $\mathcal{M}f_N(1/2+it_0) = \sqrt{N}$. Hence (24) cannot be extended to a uniform pointwise bound on the critical line. A boundary L^2 norm is a different statement from bounded evaluation at one boundary point.

8.2. **Bounds conditional on off-line zeros.**

Proposition 8.1. *If $\zeta(\rho) = 0$ and $\beta = \Re\rho > 1/2$, then for every M,*

$$D_M^2 \ge \frac{2\beta-1}{|\rho|^2}.$$

More generally, for distinct such zeros $\rho_1,\dots,\rho_k$, put $K_{ij} = 1/(\rho_i + \overline{\rho_j} - 1)$ and $v_i = 1/\rho_i$. Then $D_M^2 \ge v^*K^{-1}v$.

Proof. For any residual $f = 1 - \sum c_n g_n$, its Mellin transform is

$$F(s) = \frac{1}{s} + \frac{\zeta(s)}{s}\sum_{n=2}^{M} c_n(n^{-s} - n^{-1}).$$

Thus $F(\rho) = 1/\rho$. Applying (24) proves the single-zero bound and taking the infimum over c preserves it.

For several zeros, the evaluation representers have Gram matrix K. They are linearly independent: a vanishing linear combination of their kernels would be a rational function with distinct poles and therefore have every coefficient zero. The least norm interpolant with evaluation vector v has squared norm $v^*K^{-1}v$, as follows by projecting onto their span and solving its finite Gram system. Every residual has those evaluations, so its norm is at least that quantity. This proves the second assertion. □

The proposition neither asserts an off-line zero nor excludes one. It gives a rigorous obstruction if such a zero is supplied. In particular it provides no missing upper bound that would force $D_M \to 0$.

9. Filtering a sequence of finite approximation errors

The previous version introduced a Kalman filter for noisy evaluations of approximation errors. Such filtering is an estimation problem; it cannot establish an unproved limit of the exact sequence. We give the corrected finite formulas, with assumptions stated explicitly.

9.1. **Fixed gain.** Let $0 < K \le 1$, $r = 1 - K$, and suppose observations are $z_m = d_m + \varepsilon_m$. Define

$$\widehat{d}_m = r\widehat{d}_{m-1} + Kz_m, qquade_m = \widehat{d}_m - d_m.$$

For deterministic d_m and deterministic starting value,

$$e_m = r^m e_0 + r\sum_{j=1}^{m} r^{m-j}(d_{j-1} - d_j) + K\sum_{j=1}^{m} r^{m-j}\varepsilon_j. \tag{26}$$

This follows by substituting the recursion and iterating it; d_0 is the target value associated with the initial estimate.

Assume the noises are independent, centered and sub-Gaussian with variance proxy σ^2, meaning $\mathbb{E}e^{t\varepsilon_j} \le e^{\sigma^2 t^2/2}$ for all real t. Independence and multiplication of moment-generating functions show that the final sum in (26) is sub-Gaussian with proxy

$$\sigma^2 K^2 \sum_{j=0}^{m-1} r^{2j} = \frac{\sigma^2 K}{2-K}(1 - r^{2m}). \tag{27}$$

If the target is constant and the noises are independent with actual variance $\sigma^2 > 0$, the variance tends to $\sigma^2 K/(2-K) > 0$. Thus a fixed positive gain does not yield vanishing mean-square error in this model.

If $|d_j - d_{j-1}| \le \Delta$, its deterministic bias satisfies

$$|\mathbb{E}e_m| \le r^m |e_0| + \frac{r\Delta(1 - r^m)}{K}.$$

Combining this with (27) gives a complete finite mean-square and tail estimate. For example, with probability at least $1-\delta$, the absolute stochastic part is at most $\sqrt{2\sigma^2 K(1 - r^{2m})\log(2/\delta)/(2-K)}$.

9.2. **The random-walk gain and a vanishing-gain alternative.** In the scalar random-walk model with process variance $Q > 0$ and observation variance $R > 0$, a steady posterior variance P satisfies

$$K = \frac{P + Q}{P + Q + R}, \qquad P = RK.$$

Eliminating P yields $RK^2 + QK - Q = 0$. Its unique root in $(0, 1)$ is

$$K = \frac{\sqrt{Q^2 + 4QR} - Q}{2R}. \tag{28}$$

The underlying filtering framework is due to Kalman [5]; the scalar algebra above is included to fix the normalization.

If instead the gain is $1/m$, induction gives $\widehat{d}_m = m^{-1}\sum_{j=1}^{m} z_j$. Under the same independent noise assumption, its variance is at most σ^2/m. If the target sequence satisfies the independently established estimate $|d_j - d| \le Cj^{-\alpha}$, then

$$|\mathbb{E}\widehat{d}_m - d| \le \frac{C}{m}\sum_{j=1}^{m} j^{-\alpha}.$$

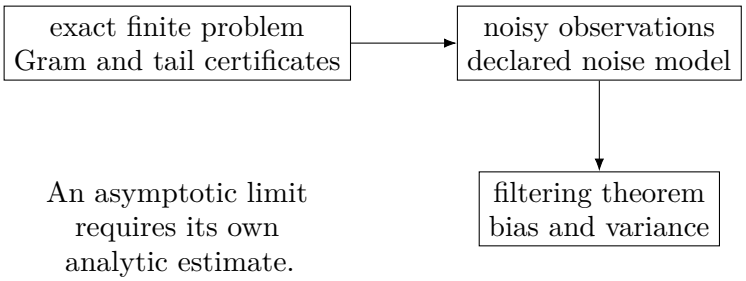


FIGURE 3. Exact approximation and statistical estimation have different inputs. A filter controls observation error under a noise model.

This is $O(m^{-\alpha})$ for $0 < \alpha < 1$, $O(\log m/m)$ for $\alpha = 1$, and $O(m^{-1})$ for $\alpha > 1$. The assumption on the exact target sequence is indispensable. Applying a filter to numerical values of D_M does not prove that assumption or RH.

10. Exact verification and limits of the results

The standard-library program `checks/verify.py` performs exact rational integration of ordinary sawtooth products, checks the GCD formula, verifies the divisor factorization and the normal equations, and compares the exact optimum with (2). It also checks the reciprocal step matrix and determinant for a finite range of M, constructs certified logarithm intervals, and bounds selected rational residuals by (22).

These finite checks support the stated identities but do not replace their proofs for arbitrary M. The graph uses only values generated from the proved ordinary-dilate formula. No synthetic experimental dataset, numerical zero exclusion or full proof-assistant audit is claimed.

The ordinary problem is solved exactly and unconditionally. The reciprocal problem has finite positive-definiteness and rigorous residual certificates, but the limit in (5) remains unproved. Classical identities and cited RH criteria are credited; the present calculations are not presented as a resolution of RH or as a certified new major conjecture closure.

Appendix A. A direct rational integration algorithm

To integrate $u_j u_k$, sort the finite set

$$\{0, 1\} \cup \{a/j : 1 \le a < j\} \cup \{b/k : 1 \le b < k\}.$$

On an open interval (α, β) in this partition, the floor values $A = \lfloor jx \rfloor$ and $B = \lfloor kx \rfloor$ are constant. The product is $jkx^2 - (jB + kA)x + AB$, so the integral is exactly

$$\frac{jk}{3}(\beta^3 - \alpha^3) - \frac{jB + kA}{2}(\beta^2 - \alpha^2) + AB(\beta - \alpha).$$

All terms are rational. This gives an independent finite computation of Lemma 3.1, with no Fourier truncation or quadrature.

Appendix B. Determinants, optimization and conditioning

For any real positive-definite Gram matrix A and target vector b,

$$\|f - \textstyle\sum c_j u_j\|^2 = \|f\|^2 - b^T A^{-1} b + (c - A^{-1}b)^T A (c - A^{-1}b).$$

This identity proves uniqueness of the minimizer and its exact error. It does not require estimating a condition number. When a floating-point approximation $\widetilde{c}$ is used instead, the excess squared error is exactly $(\widetilde{c} - c)^T A (\widetilde{c} - c)$.

For the reciprocal matrix, the positive tail in (18) shows $B \succeq B^{(Q)}$ in Loewner order. Entrywise error bounds alone would not imply that ordering. Conversely the tail's operator norm is at most its trace, hence at most $(M-1)/(Q+1)$. If $B^{(Q)}$ has a certified lower eigenvalue $\lambda > 0$, then

$$\|(B^{(Q)})^{-1} - B^{-1}\| \le \frac{M-1}{\lambda^2(Q+1)},$$

by the resolvent identity and $\|B^{-1}\| \le 1/\lambda$. This is a finite error estimate, not an asymptotic density theorem.

Appendix C. A rational logarithm certificate

For $x \ge 1$, write $t = (x-1)/(x+1)$. The identity $\log x = \log(1+t) - \log(1-t)$ follows by solving for x. Integrating the geometric series for $2/(1-t^2)$ proves (23). After K terms,

$$0 \le \log x - 2\sum_{k=0}^{K-1} \frac{t^{2k+1}}{2k+1} \le \frac{2t^{2K+1}}{(2K+1)(1-t^2)}.$$

For an integer n, choose a with $1 \le n/2^a < 2$. Apply the formula to 2 and to $n/2^a$; then use $\log n = a\log 2 + \log(n/2^a)$. All quantities in the finite sums and tail bounds are rational. The terms have $0 \le t \le 1/3$, so the tail decreases geometrically. For a signed coefficient c_n, the lower and upper logarithm endpoints must be interchanged when the coefficient multiplying the logarithm is negative.

Appendix D. Revision record

The prior version [6] discussed rank-one collapse, sawtooth inner products, Mellin transforms, Hardy bounds and filtering. All five topics are retained, but their mathematical relations have been corrected. The ordinary Gram formula is exact; its inverse, least-squares coefficients and mean constant are derived. Reciprocal functions now have separate notation, a proved step representation, a finite determinant certificate and a rational tail bound.

The previous zeta-dependent kernel and critical-line pointwise bound are replaced by (25) and the explicit boundary counterexample. The fixed-gain decay claim is replaced by the exact variance floor, and the scalar Riccati root is corrected. Claims depending on unverified numerical data or an unproved limiting estimate have been removed. The stronger title refers to the fully derived finite inverse, not a claimed solution of RH.

Declarations and availability

Competing interests. The authors declare no competing interests. **Funding.** No specific funding is reported in the supplied source. **Ethics.** The work uses no human participants or animals. **Authorship.** The author order and corresponding author of the published version are preserved. Affiliations are reproduced from the supplied sources; Pallab Nandi's IISER association is historical. **Computational and editorial assistance.** AI assistance was used for source review, derivation checks, code and preparation of this revision. It is not an author. The provenance file distinguishes the analytic proofs from the finite computational checks. No Lean formalization is claimed. **Acknowledgements.** The presentation builds on the work of Nyman, Beurling, Báez-Duarte, Smith and Kalman, and on the cited GCD sum literature. No endorsement by these researchers is implied.

Repository and release. `https://github.com/creelie/nyman-beurling-approximation`. Release v3.0.1.

Permanent work DOI. 10.5281/zenodo.24054093. This DOI identifies the work across archived releases; the version-specific DOI is recorded on the linked archive page.

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